

IQ Modulators with Two Segments using InP/SOI Chip-on-Wafer Bonding Process for Optical DAC

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Abstract: We fabricated IQ modulators consisting of two segments using InP/Si chip-on-wafer bonding. Electrical/optical 3-dB bandwidth is approximately 60 GHz ($V\pi$: 5.4 V). The symbol rate up to 64 GBaud is demonstrated in 16QAM operation using optical DAC.

1. Introduction

In recent years, the exponential growth in global data traffic has driven the demand for high-capacity, energy-efficient, and compact optical components in optical fiber communication systems. To cope with these demands, photonic integrated circuits are a key technology, and various approaches, such as InP, thin-film lithium niobate (TFLN), and silicon photonics (SiPh), have been developed up to now. However, achieving the aforementioned requirements has become increasingly challenging, especially towards multi-Tb/s era, modulator devices based on single materials have tough issues.

InP is well-suited for high-speed and high-efficiency modulation [1], utilizing its superior electro-optic (EO) properties. However, InP has limitations in achieving compact device layouts due to its large bending radius in passive waveguides. Hence it is not suitable for the high-density integration. On the other hand, SiPh offers the advantage of tight bending radii, enabling compact passive waveguide layouts for high-density integration. Despite this benefit, SiPh relies on the carrier plasma effect for modulation [2], resulting in a tradeoff between optical losses and modulator performance. The carrier plasma effect brings about the increase of optical losses with increasing carrier doping for high modulation efficiency and wide bandwidth operation. Hence, SiPh modulators are challenging to obtain high modulation efficiency and wide bandwidth operation for multi-terabit applications.

To overcome these limitations, the heterogeneous integration platform on an SOI wafer has emerged as a promising solution, and high-speed, low-voltage MZMs utilizing EO polymers [3] and thin-film lithium niobate [4] have been reported as research efforts. Additionally, it has been revealed that the direct bonding process of InP chips onto SOI wafers [5,6] is very effective for fabricating wavelength-tunable lasers [7] and high-speed modulators [8]. This bonding technology enables high-density integration of multi-functional devices, including lasers and modulators, while leveraging the advantages of InP-based QW structures.

In terms of EO signal conversion scheme to reduce power consumption, optical digital-to-analog converter (oDAC) is a promising approach [9]. The concept of oDAC is illustrated in Fig. 1. Conventionally, DACs and linear drivers were required to generate multi-level electrical signals in Fig. 1(a). However, the use of binary CMOS drivers allows to significantly reduce the power consumption in Fig. 1(b). By dividing the electrodes in the modulator and applying binary signals to each segment, multi-level optical signal generation becomes achievable.

As shown in Fig. 1(b), the block diagram represents the left segments correspond to the most significant bit (MSB) and the other segments correspond to the least significant bit (LSB). The process of generating a composite 16QAM signal using MSB and LSB is illustrated in Fig. 2. By applying QPSK signals to the MSB and LSB, the modulator generates a composite 16QAM signal. In this configuration, the MSB performs intensity modulation with twice the strength of the LSB. Divided electrode design allows the achievement of both low half-wave voltage ($V\pi$) and wide EO bandwidth. SiPh-based oDAC implementations have demonstrated the symbol rates of up to 50 GBaud for 16QAM signals [9]. However, further performance improvements can be realized by utilizing InP-based quantum well (QW) structures in the modulation section.

In this study, we successfully fabricate the IQ modulator designed for oDAC applications through the InP/SOI chip-on-wafer (CoW) bonding process for the first time. This device utilizes the high modulation efficiency of InP QW with the miniaturization capability of SiPh, addressing the critical limitations of single-material platforms. Consequently we present 16QAM operation at 64 GBaud for an InP-based IQ modulator integrated onto a SiPh

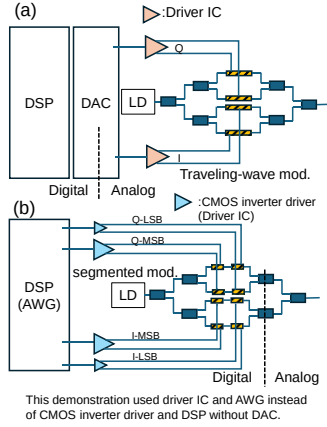


Fig. 1: Transmitter with (a) conventional architecture and (b) optical DAC architecture.

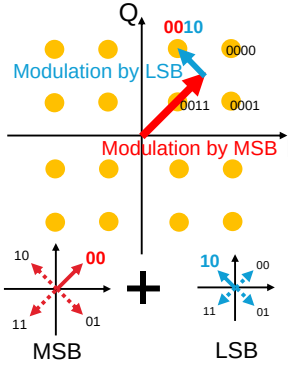


Fig. 2: Example of 16QAM modulation using MSB and LSB

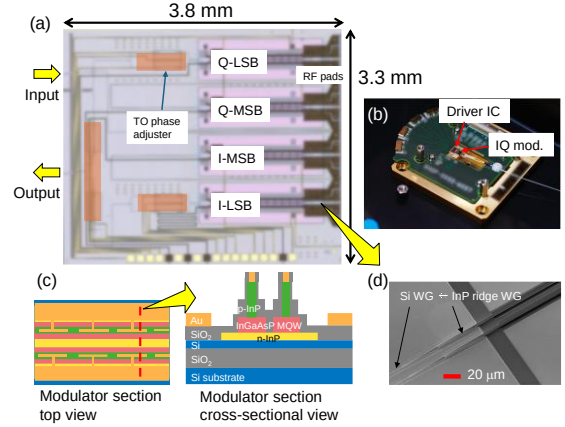


Fig. 3: (a) Photomicrograph of IQ modulator (mod.) with two segments using InP/SOI CoW bonding. (b) Photograph of IQ mod. with driver IC. (c) Schematic images of top and cross-sectional views in modulator section. (d) SEM image of optical transition structures between InP and Si waveguides (WGs).

platform compatible with oDAC technology assembling driver circuits.

2. Design and characteristic of IQ modulator with two segments using InP/SOI CoW bonding

A photomicrograph of the fabricated IQ modulator comprising two segments, is shown in Fig. 3(a). The modulator consists of four RF modulation sections (I-LSB, I-MSB, Q-LSB, Q-MSB), which are based on the heterogeneously integrated structure utilizing the InP/SOI CoW bonding process. Phase-adjustment electrodes are implemented as heaters positioned along Si waveguides. Fig. 3(b) represents the IQ modulator implemented with driver circuits. The driver ICs amplify electrical signals to effectively drive each RF modulation sections. The schematic shown in Fig. 3(c) illustrates the RF modulation sections, indicating both top and cross-sectional views. The modulation design incorporates capacitively loaded electrodes to optimize RF impedance adjustment. The filling factor (FF) configuration in RF electrodes provides flexibility in achieving desired RF impedance values. Furthermore, vertical optical confinement is achieved by the multiple quantum-well (MQW) core layer, while horizontal optical confinement is maintained by the upper InP ridge waveguide. Fig. 3(d) depicts a photograph of the taper structure for optical transition between the Si waveguide and InP ridge waveguide. Light transitions from the Si waveguide to the core layer of the InP waveguide through this taper structure. The taper configuration allows high-efficient optical transition through the precise formation achieved using the photolithography process with high positional accuracy after bonding [5,6]. Accordingly, this fabrication method minimizes optical coupling loss between Si waveguide and InP ridge sections.

After bonding, ridge waveguides, termination resistors, and RF electrical lines were fabricated using the InP process including dry etching, wet-chemical etching, and lift-off process. Fig. 4 depicts the measured optical power as a function of heater power for phase adjustments. The heater is used as single-phase driving. The heating power for the rotation of π phase ($P\pi$) is 20 mW. From this extinction characteristic, an extinction ratio is as high as 30 dB, that is, the 1×2 Multi-Mode Interference (MMI) coupler evenly splits the optical power between two arms in the RF modulator section. Fig. 5 shows the differential DC modulation characteristics of the RF modulation section using the quantum-confined Stark effect (QCSE) from the InP-based MQW core layer. The modulator operates at a center bias voltage of -8 V, resulting in high modulation efficiency with a $V\pi$ of 0.43 V·cm. Fig. 6 shows the EO response of the RF modulation sections. In Fig. 6(a), the EO characteristics of both LSB and MSB RF modulation sections are presented, and both sections exhibit an EO 3-dB bandwidth of approximately 60 GHz, indicating consistent performance across four MZ modulators. Fig. 6(b) shows the EO response after the integration of driver circuits. The Peaking of responses over 3 dB is observed between 40 and 50 GHz. Finally, the 16QAM operation with the symbol rate of 64 GBaud is achieved as the oDAC transmitter, and clear segregation among each symbol is successfully observed by an optical modulation analyzer as shown in Fig. 7.

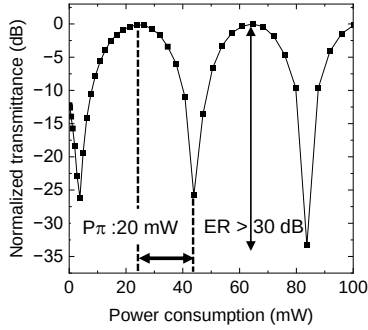


Fig. 4: Measured normalized optical power as a function of power consumption in heater for phase adjustment.

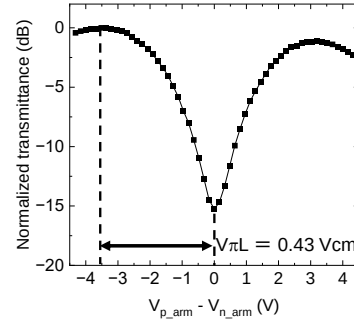


Fig. 5: Optical power versus voltage in the RF modulation section under push-pull operation, with center bias at -8 V. (An electrode length: 2 mm, FF: 60%).

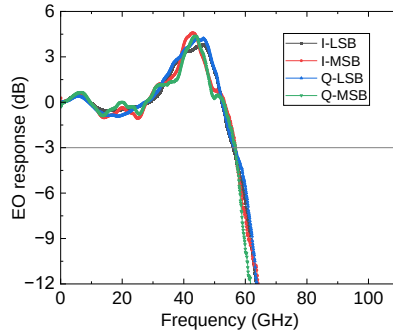
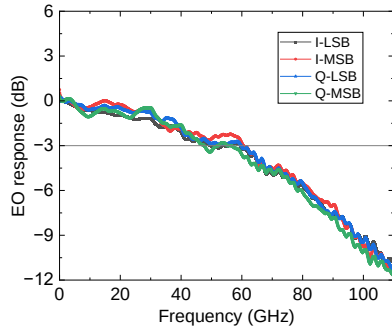


Fig. 6: Measurement result of EO response at V_π of 5.4 V (Electrode length: 1.0 mm, FF: 80%). EO response of (a) IQ modulator chip and (b) IQ modulator with driver IC.

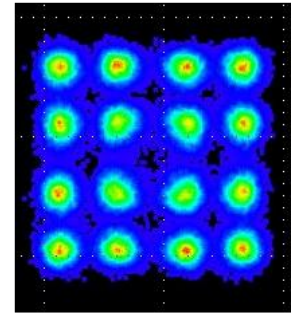


Fig. 7 16QAM constellation of optical DAC transmitter at 64 GBaud.

3. Conclusion

We have demonstrated the IQ modulator composed of InP/SOI heterogeneously integrated RF sections using the CoW bonding method. The modulator has exhibited phase modulation characteristics with the extinction ratio exceeding 30 dB and high phase modulation efficiency of 0.43 $V_\pi \cdot \text{cm}$. A wide EO 3-dB bandwidth of approximately 60 GHz has been achieved. After assembling driver ICs, the operation at the symbol rate of 64 GBaud during 16QAM modulation has been demonstrated. From these results, it has verified the vast potential of our IQ modulator using optical DAC towards the realization of energy-efficient and high-speed optical transceivers.

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